ONBOARDA OS

Product Definition

AI Compliance Operating System

Production readiness assessment, capability tiers, and strategic roadmap.

For: Investors | ARIE Finance Management | Pilot Clients | Engineering

March 2026 | v1.0

The AI Compliance Operating System

Onboarda is an AI-powered compliance operating system that reduces client onboarding from days to minutes while maintaining full regulatory traceability — purpose-built for payment institutions and corporate service providers operating under FSC, FATF, and FIAMLA frameworks.

Investors

A production-grade pilot proving that AI can collapse compliance timelines from 15 days to under 48 hours, with full audit trails and regulatory defensibility.

Pilot Clients

Secure, tested onboarding with real AI risk scoring, document verification, and compliance reporting — augmenting your team, not replacing it.

Engineering

Honest tiering: what is production-safe, what is demo-ready, and what is Phase 2 — with clear build priorities and dependencies.

The Problem → The Transformation

BEFORE ONBOARDA	AFTER ONBOARDA
Onboarding takes 10–15 days per client	Onboarding reduced to 48 hours
Manual document collection and review	Guided digital onboarding flow with auto-validation
Fragmented tools (KYC, screening, spreadsheets)	Single platform: KYC, screening, risk, compliance
Inconsistent risk assessment across officers	AI-scored risk across 5 FATF-aligned dimensions
Limited audit traceability for regulators	Full audit trail with per-agent explainability
High compliance cost per client onboarded	Lower cost per onboarding + higher throughput

Result: 80%+ reduction in onboarding time | Consistent, auditable risk decisions | Compliance officers augmented, not replaced

One Platform. Three Maturity Tiers.

PRODUCTION PILOT
Live Capabilities

- ✓ Secure, tested, operationally reliable
- ✓ Suitable for real client onboarding
- ✓ Full audit trail and regulatory defensibility
- ✓ AI-powered with human oversight

PILOT PREVIEW
Enabled Capabilities

- ✓ Visible in UI, demo-capable
- ✓ Partially implemented or hardcoded data
- ✓ Useful for stakeholder storytelling
- ✓ Not yet hardened for production reliance

PHASE 2 ROADMAP
Strategic Differentiators

- ✓ Intentionally deferred post-pilot
- ✓ Prioritised by ROI and client demand
- ✓ Architecture-ready (foundations in place)
- ✓ Build triggers: pilot validation + funding

Client Portal — Onboarding Workflow

Multi-Step Onboarding Wizard

Guided 5-step flow: company details, pre-screening questionnaire, UBO/director mapping, document upload, and final submission. All data persisted to database.

JWT Authentication

Secure client registration and login with HS256 JWT tokens stored in httpOnly cookies. Session management with issuer verification.

PEP & Ownership Declarations

Structured PEP declaration per director/UBO with public function details, ownership percentages, and nationality capture.

Draft Auto-Save

Application progress saved via API, allowing clients to resume incomplete onboarding sessions.

Document Upload with Server Validation

Real file upload to backend with extension, MIME type, magic byte, and file size validation. Path traversal protection enforced.

Dynamic Pricing Display

Risk-profile-based fee calculation shown to client before final submission, with fee acceptance gate.

Application Submission & Status Tracking

Full API integration: POST to /api/applications, real database persistence, status lookup by reference number.

Client Notification Inbox

Server-pushed notifications from compliance officers visible in portal with read/unread tracking.

Compliance Backoffice

📋 Application Review

Full case detail view with company data, director/UBO information, uploaded documents, internal notes, and decision workflow (approve / reject / escalate / request info).

📂 Case Management

Assign, reassign, and escalate applications across team members. Complete case lifecycle from submission through final decision with audit logging.

🔍 Screening Queue

Sanctions, PEP, and adverse media screening results per person. Status tracking and officer review workflow.

👤 User Management & RBAC

Four defined roles (Administrator, Senior CO, Compliance Officer, Analyst) with 18 granular permissions. User creation and role assignment.

📝 Audit Trail

Every action logged: timestamp, user, role, action type, and details. Immutable compliance record for regulatory examination.

📢 SAR Workflow

Suspicious Activity Report creation with state machine (draft → pending review → approved → filed). Database-persisted with full audit trail.

PRODUCTION PILOT AI CORE

Onboarding AI Agents — Claude-Powered Intelligence

Requires configured Anthropic API key. Production mode blocks mock fallbacks. Budget-enforced at $50/month with per-call cost tracking.

1	**Identity & Document Integrity** 66 automated checks across 16 document types. OCR text extraction, validity detection, name matching, cross-document consistency, registry verification. Per-document Pass/Warning/Fail.	claude-sonnet-4-6
2	**Corporate Structure & UBO Mapping** Ownership chain tracing, UBO identification (≥25% threshold), multi-layered entity detection, circular ownership detection, nominee/trust identification.	claude-sonnet-4-6
3	**Business Model Plausibility** Business description vs sector alignment, transaction volume benchmarking, geography vs business model logic, source of funds consistency analysis.	claude-sonnet-4-6
4	**FinCrime Screening Interpretation** Sumsub AML screening analysis: sanctions/PEP/adverse media consolidation, false positive reduction, hit severity ranking, risk narrative generation.	claude-sonnet-4-6
5	**Compliance Memo & Risk Recommendation** 5-dimension weighted composite scoring. Integrates all agent signals into structured compliance memo with risk score, level, and decision routing.	claude-opus-4-6

🧠 **Explainability Layer** Full decision trail with timestamps and confidence scores per agent. Dimension drill-down with expandable factors. Client-facing simplified view in portal.

Monitoring AI Agents — Phase 2 Build

These agents complete the continuous compliance lifecycle. Architecture-ready — supervisor framework, schemas, and pipeline routing already in codebase.

6	**Periodic Review Preparation** Automated review cycle tracking, expired document identification, change-since-onboarding summary, review memo pre-population.	NOT LIVE IN PILOT
7	**Adverse Media & PEP Monitoring** Continuous daily scans for new sanctions listings, PEP status changes, enforcement actions, adverse media alerts. Severity classification.	LIVE
8	**Behaviour & Risk Drift** Transaction pattern comparison vs onboarding profile, new jurisdiction detection, volume spike analysis, sector change identification.	NOT LIVE IN PILOT
9	**Regulatory Impact** New circular/rule detection, impacted client segment mapping, compliance action triggering, regulatory change timeline tracking.	FUTURE PHASE
10	**Ongoing Compliance Review** All monitoring signals consolidated, risk trend analysis, maintain/EDD/exit recommendation, periodic review memo generation.	NOT LIVE IN PILOT

Build trigger: Pilot validation + Tier 1 funding. Supervisor orchestration layer (contradiction detection, confidence routing) enables all monitoring agents.

Security Hardening & Platform Infrastructure

JWT Authentication

HS256 tokens in httpOnly cookies with issuer verification. Migrated from localStorage. Rate-limited login endpoints.

XSS Prevention

innerHTML replaced with textContent across all user-facing chat and display components. escapeHtml() utility enforced.

Prompt Injection Defence

SYSTEM/IGNORE/OVERRIDE pattern filtering, HTML tag stripping, 500-character input limits on all AI-bound user input.

File Upload Security

Extension whitelist, MIME type validation, magic byte verification, size limits, and path traversal protection via os.path.basename().

50+ REST API Endpoints

Full CRUD for applications, documents, users, SAR reports, monitoring, and configuration. Tornado async framework.

SQLite / PostgreSQL

Persistent relational storage with schema migrations. SQLite for dev, PostgreSQL for production. Connection pooling.

AI Budget Enforcement

Per-call cost tracking with $50/month cap. Production mode blocks mock fallbacks. Token usage logging per agent.

Audit Logging

Every API action logged: user, role, timestamp, action type, IP address. Immutable records for regulatory examination.

Current Preview — Enabled Capabilities

These features are visible in the platform and useful for demos, but rely on simulated data or partial implementations. They should not be presented as production-safe.

Regulatory Intelligence (Lite)	Document upload with metadata tagging, AI summary generation, impact mapping, and human review workflow. AI analysis uses keyword pattern matching (not Claude API). Suitable for demonstrating the concept and workflow – not for real regulatory decisions.
Company Registry Lookup	Client portal shows OpenCorporates-style company search with auto-prefill. Currently uses simulated responses when API key is not configured. Real integration available when OPENCORPORATES_API_KEY is set.
KPI Dashboards & Analytics	Backoffice shows 20+ KPI metrics across operations, risk quality, and team productivity. Current dashboards use demo data. Real analytics will be activated in Phase 2 with live database aggregation queries.
Ongoing Monitoring & EDD Pipeline	UI for monitoring alerts, periodic reviews, and enhanced due diligence. Renders hardcoded sample data. No real transaction monitoring or trigger engine.
AI Chat Assistant	Backoffice AI assistant for compliance questions. Functional when Claude API key is configured; returns mock responses otherwise.

Strategic Roadmap

TIER 1 — Highest ROI / Next Build

- Real KPI Analytics — Replace hardcoded dashboards with live database queries and trend calculations
- Agent 6: Periodic Review Preparation — Automated review cycle tracking, expired document identification
- Agent 8: Behaviour & Risk Drift — Transaction pattern monitoring, volume spike detection
- CSRF Protection & Security Audit — SameSite enforcement, CSRF tokens, penetration testing

TIER 2 — Strategic Differentiators

- AI Agent Supervisor — Orchestration layer: contradiction detection, confidence thresholds, consensus escalation
- Agent 10: Ongoing Compliance Review — All monitoring signals consolidated, risk trend analysis
- Full Regulatory Intelligence Engine — Live scraping (FSC, FATF, EU), multilingual analysis, one-click rule apply
- Multi-Jurisdiction Rule Engine — Configurable rules per jurisdiction (FSC, FCA, AMLD6, MAS)

TIER 3 — Later Enhancements

Agent 9: Regulatory Impact (Future Phase) | Data Retention & GDPR Automation | Client Self-Service Portal Expansion | Board-Ready PDF Reporting | World-Check / Dow Jones / LexisNexis Integrations

What Should NOT Be Described as Live

Credibility requires honesty. These items must not be presented as production-ready until the stated conditions are met.

KPI Dashboards	All metrics are hardcoded. Must not be described as real analytics until connected to database aggregation queries.
Ongoing Monitoring (Agents 6, 8, 10)	Agents 6, 8, 10 not live in pilot. Must not be described as active monitoring until these agents are enabled.
Regulatory Intelligence AI	Uses keyword matching, not Claude. Must not be described as AI-powered analysis until connected to real language model.
Transaction Monitoring	UI exists but no monitoring engine. Must not be described as ongoing monitoring until trigger rules and data feeds are implemented.

Recommended Messaging by Audience

INVESTORS

"Onboarda is a production-grade AI compliance operating system currently in live pilot with ARIE Finance. The platform automates end-to-end client onboarding with AI-powered risk scoring across five FATF-aligned dimensions, reducing processing time from 15 days to under 48 hours. Core capabilities — application workflow, AI agents, case management, and security infrastructure — are production-tested and audit-ready. External integrations (sanctions screening, KYC verification) are code-complete and activated upon credential configuration. Phase 2 expands into multi-jurisdiction rule engines, real-time monitoring, and a full regulatory intelligence pipeline."

PILOT CLIENTS

"Onboarda streamlines your compliance onboarding with a secure, guided workflow backed by AI risk assessment. Your clients submit applications through a branded portal, and your team reviews them through a full compliance dashboard with audit trails, case management, and decision workflows. AI agents score risk across five dimensions and generate compliance memos — but every decision remains with your team. The platform is designed to augment your compliance officers, not replace them."

ENGINEERING

"Production tier: Application CRUD, JWT auth, file validation, 5 AI agents (Claude-powered), Sumsub KYC/AML, SAR workflow, and audit trail. Preview tier: Company registry lookup (OpenCorporates) code-complete but falls back to mock without credentials. KPI dashboards use demo data. Regulatory Intelligence uses keyword matching. Tier 1 priorities: real analytics queries, CSRF protection, Agent 6 (periodic reviews), Agent 8 (risk drift), and integration test suite."

Business Model

Implementation Fee

$20K – $50K

- ✓ Platform setup & configuration
- ✓ Integration with existing systems
- ✓ Custom rule configuration
- ✓ Staff training & onboarding

Monthly Platform Fee

$2K – $5K / institution

- ✓ Full platform access
- ✓ AI agent processing
- ✓ Ongoing support & updates
- ✓ Regulatory rule updates

Usage-Based Pricing

$20 – $100 / onboarding

- ✓ Tiered by complexity
- ✓ Volume discounts available
- ✓ Includes all 5 AI agents
- ✓ Full audit trail per case

Why This Works

Clients save $5K–$15K per onboarding in manual review costs, reducing processing from 15 days to under 48 hours. At 50+ onboardings per year, the platform pays for itself within months. Premium monitoring modules provide additional recurring revenue.

COMPETITIVE LANDSCAPE

Why Onboarda Wins

Fenergo

POSITIONING

Enterprise-grade compliance

WEAKNESS

Slow implementation (6–12 months)

WEAKNESS

High cost ($500K+ annual)

WEAKNESS

Heavy customisation required

GAP

Not built for MEA PSPs/CSPs

ComplyAdvantage

POSITIONING

Strong sanctions/AML data layer

WEAKNESS

Data-focused, not workflow-driven

WEAKNESS

No end-to-end onboarding flow

WEAKNESS

Limited decision engine

GAP

No compliance memo generation

Onboarda

STRENGTH

Fast deployment (weeks, not months)

STRENGTH

AI-powered decision layer (5 agents)

STRENGTH

End-to-end workflow with audit trail

STRENGTH

Full explainability & auditability

STRENGTH

Purpose-built for PSPs/CSPs in MEA

ONBOARDA

Built for Credibility. Built for Compliance. Built for Scale.

The pilot proves the thesis. Phase 2 expands the platform.

1 **Pilot Validation**
Real client onboarding with production capabilities

2 **Integration Activation**
Live sanctions, KYC, and registry APIs

3 **Phase 2 Build**
Supervisor, monitoring, multi-jurisdiction engine